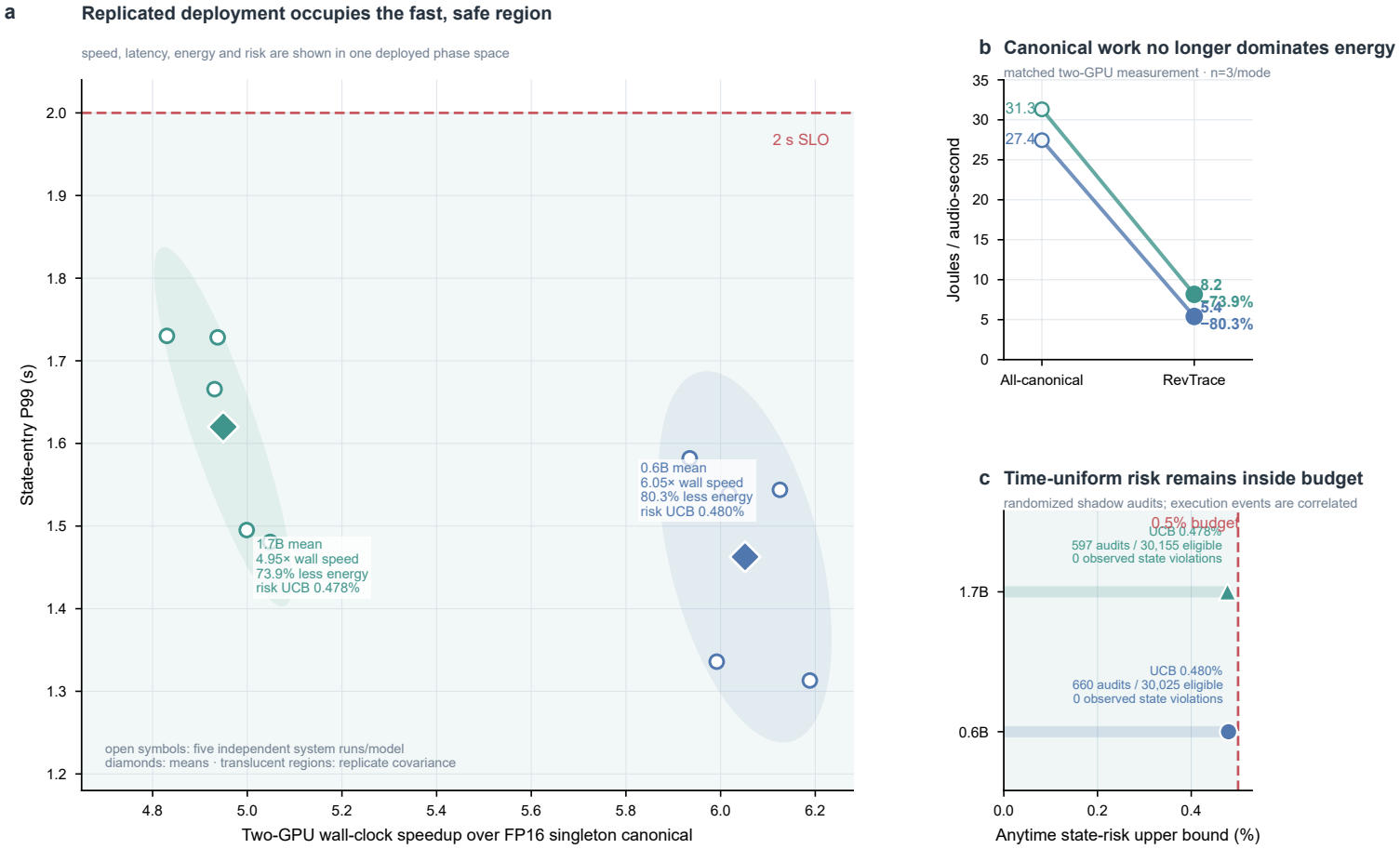

a
Replicated deployment occupies the fast, safe region
speed, latency, energy and risk are shown in one deployed phase space
2 s SLO
State-entry P99 (s)
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.7B mean
4.95× wall speed
73.9% less energy
risk UCB 0.478%
0.6B mean
6.05× wall speed
80.3% less energy
risk UCB 0.480%
open symbols: five independent system runs/model
diamonds: means · translucent regions: replicate covariance
4.8
5.0
5.2
5.4
5.6
5.8
6.0
6.2
Two-GPU wall-clock speedup over FP16 singleton canonical
b
Canonical work no longer dominates energy
matched two-GPU measurement · n=3/mode
Joules / audio-second
35
30
25
20
15
10
5
0
31.3
27.4
8.2
5.4
−73.9%
−80.3%
All-canonical
RevTrace
c
Time-uniform risk remains inside budget
randomized shadow audits; execution events are correlated
0.5% budget
UCB 0.478%
597 audits / 30,155 eligible
0 observed state violations
1.7B
UCB 0.480%
660 audits / 30,025 eligible
0 observed state violations
0.6B
0.0
0.2
0.4
Anytime state-risk upper bound (%)